



Plant Protection Association of India

Plant Protection Association of India Golden Jubilee Celebrations 1972 - 2022
PJTAU Campus, Rajendranagar, Hyderabad 500030, Telangana, India

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Date: August 4, 2026

To,

Shri Muktanand Agrawal, The Joint Secretary (Plant Protection),

Room No. 227, Ministry of Agriculture and Farmers Welfare,

Government of India, Krishi Bhawan, New Delhi – 110001.

Email: muktanand.ias08@nic.in, jspp-dac@gov.in

Subject: Comprehensive Economic Impact Assessment, Technical Evaluation, and Regulatory Suggestions on the Draft Notification for the Prohibition of Paraquat Dichloride

Dear Sir,

The Plant Protection Association of India (PPAI) respectfully submits this comprehensive representation on behalf of our scientific community, associated Farmer Producer Organisations (FPOs), and agricultural stakeholders regarding the draft notification proposing a nationwide ban on the import, manufacture, sale, and use of Paraquat Dichloride 24% SL.

While PPAI fully supports the Ministry's intent to eliminate hazardous risks from the agricultural ecosystem, an abrupt, absolute ban on this molecule creates a severe structural and financial vacuum. At a time when Indian smallholders are already managing an extreme "**Super El Niño**" crisis—marked by severe rainfall deficits and rising rural economic stress—removing this foundational, low-cost weedicide without an immediate alternative will compromise crop yields and accelerate input inflation.

We are herewith presenting critical agronomic, economic, and scientific ground realities, followed by a balanced, phased regulatory compromise for your consideration.

I. The Grassroots Economic Reality (FPO & Field Perspective)

- **Severe Input Cost Inflation:** Paraquat remains the most economical broad-spectrum contact herbicide available to Indian smallholders, costing roughly ₹450 to ₹600 per acre. The closest chemical contact match under evaluation, Glufosinate-Ammonium, costs between ₹1,200 and ₹1,500 per acre. Forcing a sudden shift will instantly increase weed management expenses by 150% to 200%, placing an unsustainable financial burden on marginal farmers.
- **Worsening Labour costs & Volatile Weather:** Finding manual labour during the peak sowing window is highly difficult and expensive. Paraquat provides an instant, labour-saving knockdown effect. Furthermore, under current climate uncertainties, dry spells are frequently broken by sudden convective downpours. Unlike systematic herbicides that require 6 to 8 hours of dry weather, Paraquat is completely rain-fast within 30 minutes, allowing farmers to clear fields effectively right before rain episodes occur.
- **Threats to Conservation Agriculture:** Paraquat serves as the agronomic anchor for zero-tillage and conservation agriculture across India. It clears heavy weed cover within 48 hours without disturbing the soil architecture. An immediate ban will force farmers back to intensive mechanical tractor-tilling, which accelerates soil moisture evaporation, increases diesel consumption, and degrades soil health.

II. Technical Evaluation of Toxicity vs. Exposure Pathways (Scientific Perspective)

- **Analyzing the Exposure Route:** Rigorous toxicological data shows that Paraquat's hazard profile is linked to *intentional or accidental oral ingestion* rather than regular, standard occupational spraying. Its dermal absorption rate through intact human skin is remarkably low (<1%). Occupational issues stem entirely from low-quality, knapsack sprayers and a lack of basic Personal Protective Equipment (PPE).



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- **Global Precedent (The "Restricted Use" Model):** A total ban overlooks successful global regulatory frameworks. In advanced agricultural systems like the United States (US-EPA) and Australia (APVMA), Paraquat remains a legal, highly vital tool for conservation tillage. These nations have successfully mitigated public health risks not by banning the molecule, but by classifying it strictly as a **"Restricted Use Pesticide" (RUP)**—its handling entirely limiting to licensed, certified applicators.

III. Constructive Policy Recommendations and Phased Roadmap

To balance the safety of the farming community with the economic security of Indian agriculture, PPAI strongly urges the Ministry to adopt a **stricter restriction framework** rather than an absolute prohibition:

1. **Enforce an 18 to 24-Month Transition Window:** Defer a complete ban for a minimum of 1.5 to 2 years. This grace period will prevent sudden retail supply chain shocks, allow FPOs to adjust seasonal crop budgets, and give the Central Insecticides Board & Registration Committee (CIB&RC) time to fast-track the testing and registration of lower-toxicity chemical alternatives (such as Glufosinate-Ammonium or Carfentrazone-ethyl combinations).
2. **Implement a "Certified Spray Applicator Only" System:** Shift the regulatory classification of Paraquat from open retail to "Strictly Restricted." Restrict its sale and field application exclusively to government-licensed Custom Hiring Centres (CHCs), FPOs, and certified professional operators equipped with closed-delivery, low-drift nozzles and mandatory PPE compliance.
3. **Mandate Enhanced Manufacturing Safeguards:** Force all technical manufacturers to modify the chemical formulation by significantly increasing the concentration of safeguarding emetics (vomit-inducing compounds), strong stenching/olfactory alert agents, and permanent blue dyes. This ensures that accidental or intentional ingestion is immediately rejected by the human body.
4. **Fund an Emergency Antidote Taskforce:** Direct a portion of corporate agrochemical cess and government research funds to the Indian Council of Medical Research (ICMR) to initiate a fast-track, time-bound research grant aimed at identifying an effective medical antidote or advanced adsorption therapy protocol.
5. **Subsidize Mechanical Alternatives:** To protect conservation agriculture, the Ministry should divert a portion of subsidies toward providing heavy **90% financial grants for FPOs** to procure mechanical power weeders, intra-row chiselers, and laser weeders under the Sub-Mission on Agricultural Mechanization (SMAM).

Way Forward

An absolute, overnight ban is an administrative mechanism that will heavily penalize vulnerable smallholders and disrupt the progress of conservation farming in India. Transforming Paraquat into a regulated, professionally applied restricted substance represents the most scientifically balanced and economically secure path forward for our nation.

The PPAI stands ready to undertake needful studies on the ground to generate detailed bio-efficacy reports and cooperate with the Ministry's Registration Committee to hammer out the operational guidelines for this restricted-use framework.

I on behalf of the Executive Committee of PPAI, thank you profusely for your dedication to Indian agricultural science and the welfare of our farming community.

Sincerely Yours,

Dr. Sarath Babu Balijepalli

President, Plant Protection Association of India (PPAI)